UNIT - 1

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Concept of Watershed | Introduction to Watershed Management | Stakeholders and relative importance | Policies and decision making | Practices in arid & semi-arid regions | Short term & long term strategic planning | Types & sources of pollution | Environmental guidelines for water quality | Perspective on recycle and reuse

1. Concept of Watershed

Watershed: the area of land from which all rain water (surface runoff) drains to a common outlet such as a stream, river, lake, reservoir or the sea. It is bounded by a high ridge line called the water divide.

Other names: catchment, drainage basin, river basin (in the USA: watershed; in the UK: catchment).

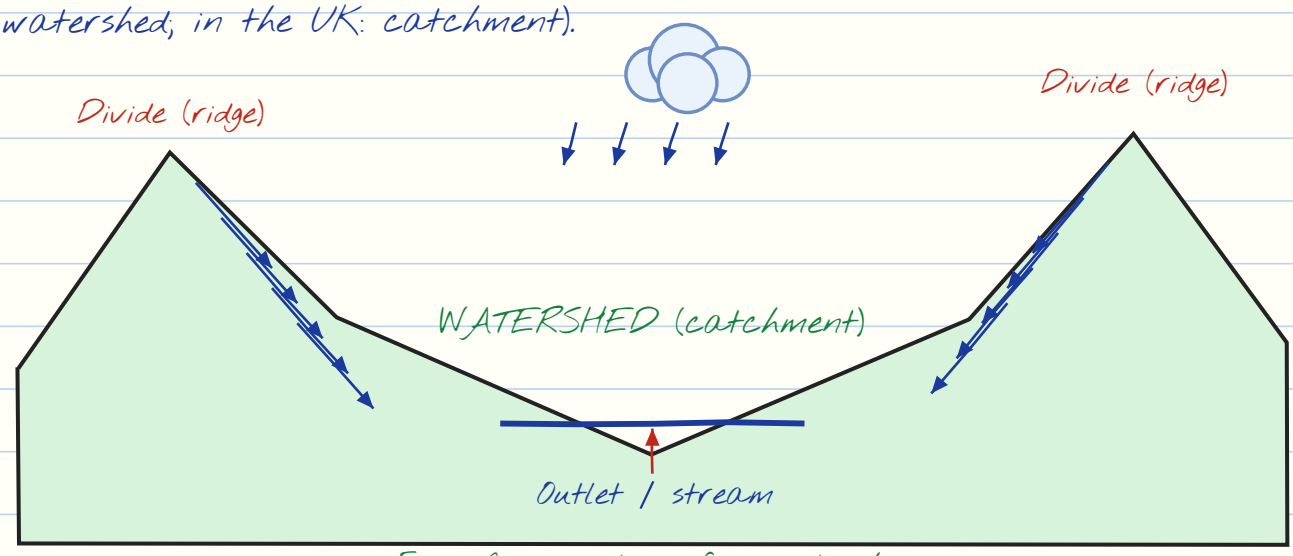


Fig 1: Cross-section of a watershed

Components / Characteristics

* **Divide:** the ridge line that separates two watersheds. Rain falling on either side flows to a different outlet.

* Outlet (pour point): the lowest point where all runoff leaves the watershed.
* Drainage network: main stream + tributaries (streams are ordered 1st, 2nd, 3rd ... by Strahler method).
* Land use: forest, agriculture, grazing land, wasteland, settlements. Land use controls runoff and erosion.
* Physical features: area, shape, slope, relief, soil type, geology, vegetation cover.

Classification by size (AIS and LUS, India)



Fig 2: Hierarchy of watersheds (by area)

* Hydrologically each watershed behaves as an open system: Input = rainfall; Output = runoff, ET, percolation, storage change.

$$P = Q + ET + dS$$ (water balance)

where P = precipitation, Q = runoff, ET = evapotranspiration, dS = change in storage (soil moisture + groundwater).

Why the watershed is the best planning unit?

1. Natural hydrological unit: effect of any action upstream (land use, dam, pollution) is felt downstream at the outlet.
2. Links land, water, vegetation and people in one system - an integrated approach is possible.
3. Upstream-downstream linkages can be studied: erosion upstream means siltation downstream.

4. Easy to monitor: measuring flow and sediment at one outlet gives the effect of all treatments.
5. Suitable for community participation because a micro-watershed generally covers 1 to 3 villages.

2. Introduction to Watershed Management

Watershed Management: the process of formulating and carrying out a course of action involving manipulation of the natural system of a watershed to achieve specific objectives - while protecting soil, water and vegetation and improving the livelihood of people living in it.

> Definition in short:
> Watershed management is NOT just soil and water conservation. It is the integrated use of land, water and vegetation (and also human and livestock resources) on a drainage-basin basis for optimum and sustained production with minimum hazard to natural resources.

Objectives

* Control runoff and soil erosion; conserve rain water in situ (where it falls).
* Increase infiltration, recharge groundwater, raise water table in wells.
* Improve soil fertility and increase crop, fodder, fuel and timber production.
* Reduce siltation of tanks, reservoirs and flood peaks; protect downstream areas.
* Ensure sustainable livelihood, reduce poverty, create employment in rural areas.
* Maintain ecological balance - biodiversity, water quality, climate resilience.

Principles of watershed management

* Utilise land according to its capability (land capability classes I to VIII).
* Protect fertile land from degradation and put every field under vegetative cover.
* Conserve as much rain water as possible where it falls (in situ).
* Drain out excess water safely at non-erosive velocity.
* Store excess water in ponds/check dams for supplemental irrigation.
* Treat the watershed ridge-to-valley: upper reach -> middle reach -> lower reach.
* Build capacity of people, make users the owners and managers of the work.

Scope / Components of the programme

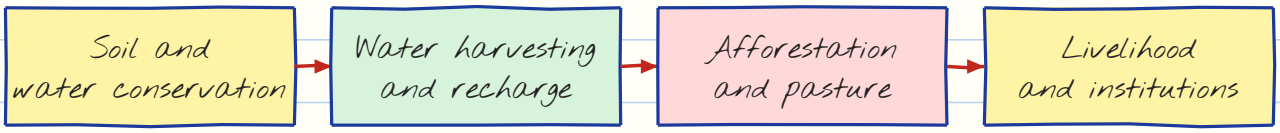


Fig 3: Main components

Component	Typical measures
Land treatment	contour bunds, terraces, trenches, gully plugs, vegetative barriers
Drainage line	check dams, loose boulder structures, gabions, farm ponds
Biomass	agro-forestry, silvi-pasture, horticulture, fodder
Livelihood	SHGs, dairy, micro-enterprise, training

3. Different Stakeholders and Their Relative Importance

Stakeholder: any person, group or institution that affects, or is affected by, the watershed and its management.

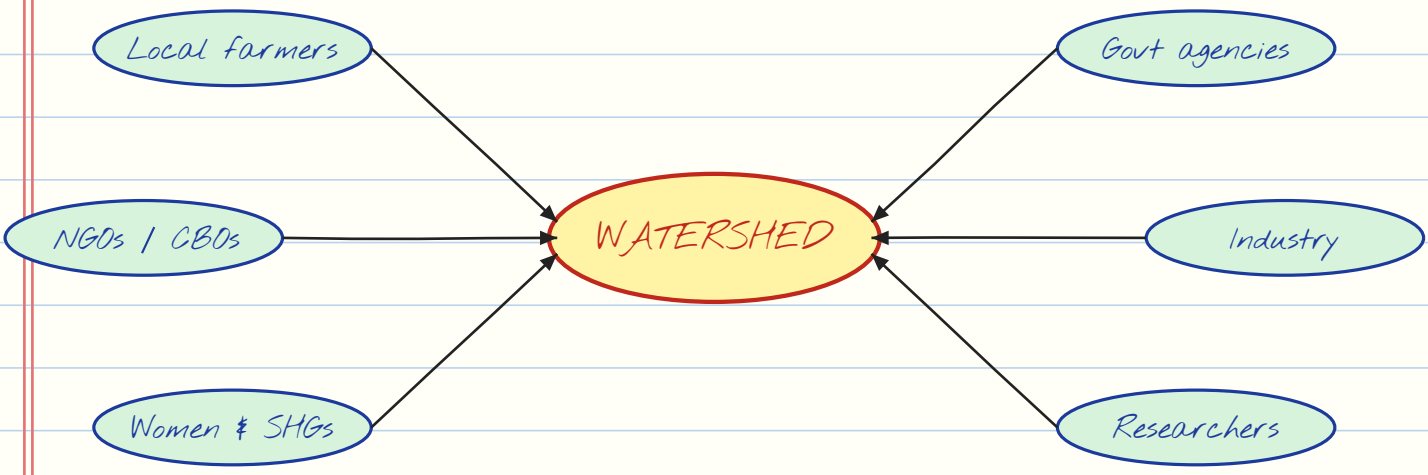


Fig 4: Stakeholders around a watershed

Main groups

* Primary stakeholders: farmers, landless labour, women, livestock keepers, tribal and forest dwellers - directly depend on the land and water; they bear the cost and get the benefits.
* Secondary stakeholders: Gram Panchayat, line departments (agriculture, forest, irrigation, rural development), banks, NGOs, research institutes, extension agencies.
* Tertiary / external: downstream users, urban water supply, industries, tourists, policy makers, funding agencies.

Relative importance (influence - interest)

Group	Interest	Influence	Role / need
Small farmers, women	Very high	Low (needs support)	Primary users, to be empowered and given a voice
Large landholders	High	High	Can block or help, benefits must be shared fairly
Government depts	Medium	Very high	Funds, technology, rules, monitoring
NGOs / CBOs	High	Medium	Facilitate, mobilise, train, bridge gap
Industry / urban users	Medium	High	Water demand, pollution, can fund CSR
Researchers	Medium	Low-Med	Data, models, impact assessment

Remember:

Equity rule: priority should go to the high-interest-low-influence group (poor, women, landless). If they are ignored, the project fails after the funding ends. Stakeholder analysis = identify -> map -> engage -> resolve conflicts.

Conflicts arise: upstream vs downstream, farmers vs graziers, landed vs landless. Solved through Watershed Committee, User Groups, transparent social audit and benefit sharing.

4. Watershed Management Policies and Decision Making

Policy framework in India

* National Water Policy (1987, 2002, 2012): water as a national resource, river basin as unit, priority to drinking water, then irrigation, hydropower, industry.
* Integrated Watershed Management Programme (IWMP, 2009-10): merged DPAP, DDP, IWDP, later in PMKSY - Watershed Development component (2015).

* Common Guidelines for Watershed Development Projects 2008 (revised 2011): cluster approach, 4-7 year project, 5-8 per cent for capacity building, 1000 ha per project minimum.
* Other: MGNREGA (works on water conservation), National Mission for Sustainable Agriculture, Jal Shakti Abhiyan, Atal Bhujal Yojana, Mission Amrit Sarovar.

Decision-making in watershed management

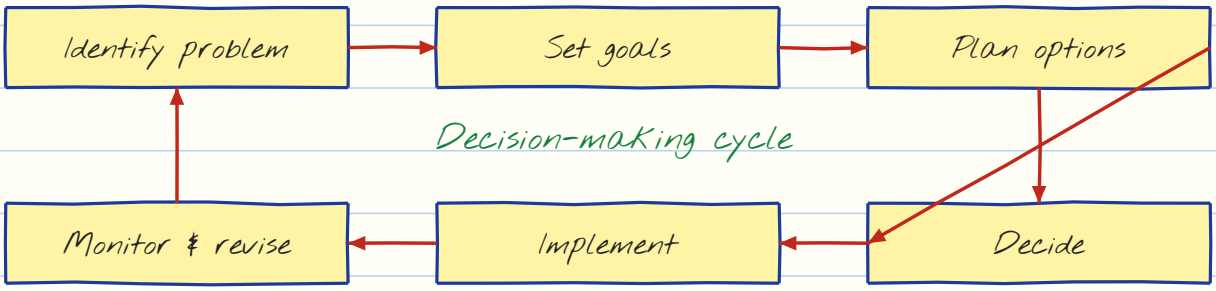


Fig 5: Participatory decision cycle

* Bottom-up: PRA (Participatory Rural Appraisal) with villagers decides problems and priorities.
* Tools: GIS and remote sensing, hydrological models, multi-criteria analysis, cost-benefit analysis, stakeholder consultation.
* Criteria for priority: runoff potential, erosion class, slope, land use, poverty, drinking-water scarcity, groundwater status.
* Adaptive management: monitor, learn and change decision with new information.

> Principle:
> Good decisions are: participatory, transparent, science-based, equitable, economically viable and environmentally sustainable.

5. Practices in Arid and Semi-arid Regions

Arid: rainfall below 250-400 mm/yr, very high PET; Semi-arid: rainfall 400-800 mm with erratic distribution, long dry spells and frequent droughts. About 2/3 of India's cultivated area is rainfed.

Problems

* Low and highly variable rainfall; high intensity storms cause erosion.
* Shallow, coarse, low-organic-matter soils; crusting; wind erosion in deserts.
* Low groundwater recharge, salinity, over-extraction; drought-prone livelihoods.

Management practices

* In situ moisture conservation: contour bunding, graded bunds, compartment bunds, contour trenching, broad-bed and furrow, tied ridges, mulching, conservation tillage.
* Water harvesting: farm ponds, nala bunds, check dams, percolation tanks, Khadin (Rajasthan), tankas, johads, ahar-pyne, tanka.
* Wind erosion control: shelter belts, wind breaks, sand-dune stabilisation using grasses (Cenchrus, Lasiurus) and trees (Prosopis, Acacia).
* Drought-tolerant crops: bajra, jowar, pulses, castor; contingency crop plans; short-duration varieties.
* Silvi-pasture and agro-forestry: Hardwickia, Prosopis cineraria (Khejri), Ziziphus (ber), Azadirachta (neem).
* Efficient irrigation: drip, sprinkler, protective irrigation, supplemental irrigation at critical stages.
* Contingency: fodder banks, crop insurance, grain banks, community seed banks.

> Slogan:
> Rule for dry regions: 'Catch the rain where it falls, store it, use it efficiently.'

6. Short Term and Long Term Strategic Planning

Aspect	Short term (1-3 yrs)	Long term (5-25 yrs)
Goal	quick visible benefits, build trust	sustainable productivity, ecology
Actions	gully plugs, field bunds, farm ponds, drinking water, nursery	afforestation, terracing, groundwater recharge, institutions, markets
Finance	project grants, wage labour	watershed development fund, user charges, bank credit
Output	reduced runoff, more soil moisture	raised water table, stable yields, biodiversity

* Strategic planning steps: (1) vision and objectives, (2) baseline survey and resource inventory, (3) problem analysis, (4) alternatives and priorities, (5) action plan, budget, schedule, (6) implementation, (7) monitoring and evaluation (M and E), (8) exit and sustainability.
* Phases in watershed projects: preparatory (3 yrs) -> works phase (3-4 yrs) -> consolidation / withdrawal phase.
* Short-term wins keep people interested; long-term plans safeguard the resource base. Both are linked through a Detailed Project Report (DPR).

7. Types and Sources of Pollution

Water pollution: addition of unwanted substances (physical, chemical, biological) which make water unfit for its intended use.

Point sources (pipes) vs Non-point sources (runoff)

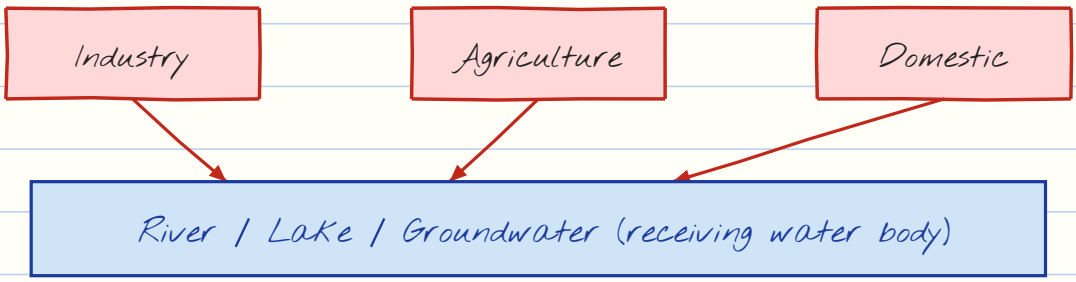


Fig 6: Pollution sources

Sources

* Point source: from a single identifiable location - industrial effluent pipe, sewage outfall, mine drainage. Easy to monitor, can be treated.
* Non-point (diffuse) source: no single point - agricultural runoff (fertiliser, pesticides), urban storm water, soil erosion. Hard to control.

Types of pollutants

Type	Examples	Effect
Organic	sewage, food waste, paper mills	high BOD, low DO, fish kill
Nutrients	nitrogen and phosphorus from fertiliser	eutrophication, algal bloom
Pathogens	bacteria, viruses, protozoa	cholera, typhoid, hepatitis
Sediment	eroded soil	turbidity, reservoir siltation
Toxic / heavy metals	Pb, Hg, Cd, As, pesticides	bio-accumulation, cancer
Thermal	power plant cooling water	reduced DO, ecosystem change
Oil / plastics / emerging	oil spills, microplastics, pharmaceuticals	toxic to aquatic life

* Watershed link: erosion + fertiliser + livestock waste + poor sanitation in the catchment decide the quality of water at the outlet.

8. Environmental Guidelines for Water Quality

Standards give the permissible limits of each parameter for a given use.

Common bodies: BIS (IS 10500), CPCB, WHO, US-EPA.

Parameter	Drinking (IS 10500)	Remarks
pH	6.5 – 8.5	acid/alkaline limit
TDS	500 mg/L (2000 if no alternative)	taste, salinity
Turbidity	1 NTU (5 permissible)	pathogens hide in particles
Hardness	200 mg/L (600)	as CaCO3
Nitrate	45 mg/L	blue baby disease
Fluoride	1.0 mg/L (1.5)	fluorosis
Arsenic	0.01 mg/L	toxic, cancer
Coliform	absent in 100 mL	faecal contamination

* CPCB classes of surface water: A (drinking after disinfection, DO > 6, BOD < 2), B (outdoor bathing, BOD < 3), C (drinking after conventional treatment, BOD < 3), D (fish and wildlife, DO > 4), E (irrigation, industrial cooling).
* Irrigation water: EC < 2 dS/m, SAR (sodium adsorption ratio) < 10 for safe use; salinity and sodium hazard classes of US Salinity Lab.
* Environmental flow: minimum flow to keep aquatic ecosystem healthy must be left in streams.
* Effluent discharge standards: BOD <= 30 mg/L for inland surface waters (CPCB general standard), COD <= 250 mg/L, TSS <= 100 mg/L.

BOD = (DO initial − DO after 5 days at 20 C) × dilution factor

9. Perspective on Recycle and Reuse

Scarcity and rising demand make water a resource that must be used again and again – 'reduce, reuse, recycle, recharge'.

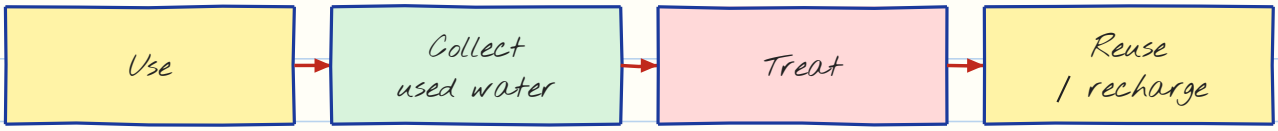


Fig 7: Reuse loop

* Reuse: using wastewater (with or without treatment) again for another purpose, e.g. treated sewage for gardens, toilets, construction, cooling.
* Recycle: treated water used again for the same purpose in a closed loop, e.g. industrial process water.
* Greywater (bath, laundry, kitchen) -> toilet flushing, gardening. Blackwater needs full treatment.
* Treated municipal wastewater (STP, constructed wetlands) -> irrigation of non-food/fodder/treecrops, industrial cooling, groundwater recharge.
* Farm level: tail-water recovery, return flow, use of drainage water in conjunctive way with fresh water.
* Rain water harvesting is also 'reuse' of natural precipitation.

Benefits and concerns

* Benefits: reduces fresh water demand and pollution load, cheap supply, nutrients (N, P) for crops, drought-proof source.
* Concerns: pathogens, salts, heavy metals, public acceptance, needs treatment standards, monitoring and health safeguards.

> Take-away:
> Wastewater is a resource, not a waste. Safe reuse = right treatment + right crop/use + right application method + monitoring (WHO multi-barrier guidelines).

Examples: Singapore NEWater, Israel (reuses > 85% of its sewage for agriculture), Nagpur STP water for power plant, Surat treated water for industries.

UNIT - II

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Syllabus Topics:

Morphometry | Soil erosion: factors, effects on fertility and land capability | Soil erosion modelling | Erosivity and erodibility | Sediment yield and sedimentation | Wetland definitions, role of water | Wetland water budgets and hydroperiod | Inflows, outflows, storage | Precipitation, runoff, evapotranspiration

1. Morphometry of a Watershed

Morphometry: the quantitative measurement and mathematical analysis of the shape, size, relief and drainage pattern of a watershed. It helps to predict hydrologic response (runoff, peak flow, erosion) even in ungauged basins.

A. Linear aspects

* Stream order (u): Strahler method - smallest unbranched streams = 1st order; when two streams of order u meet they form order u+1; a lower-order stream joining a higher order does not change the order.
* Stream number (Nu): number of streams of each order. Bifurcation ratio Rb = Nu / N(u+1). Normal range 3 - 5; high Rb (> 5) means strong structural control, elongated basin, low flood peak.
* Stream length (Lu): total length of streams of order u. Mean stream length ratio RL = Lu / L(u-1).
* Horton's laws: law of stream numbers, law of stream lengths, law of stream areas - they show geometric series relationships.
* Basin length (Lb): longest dimension of the basin parallel to main drainage line.

B. Areal aspects

Drainage density $Dd = L / A$ (Km per sq Km)

L = total length of all streams, A = basin area. High Dd: impermeable soil, sparse vegetation, steep slope -> quick runoff and high erosion. Low Dd: permeable soil, dense vegetation.

Stream frequency $Fs = N / A$ (streams per sq Km)

Form factor $Rf = A / Lb^2$

Circularity ratio $Rc = 4 \pi A / P^2$

Elongation ratio $Re = 2 \sqrt{(A/\pi)} / Lb$

Index	Low value means	High value means
Form factor Rf (max 0.78)	elongated basin, flat peak, low flood	circular basin, sharp peak, high flood
Circularity Rc (0 to 1)	elongated	circular
Elongation Re (0.6-1.0)	high relief, steep slope	low relief, flat region
Drainage density	coarse texture, permeable	fine texture, impermeable

* Constant of channel maintenance C = 1/Dd (area needed to maintain 1 Km of channel). Length of overland flow Lo = 1/(2 Dd).
* Texture ratio T = N1/P (first order streams per Km of perimeter).

C. Relief aspects

* Basin relief (H): difference in elevation between highest point and outlet.
* Relief ratio Rh = H / Lb, Ruggedness number Rn = H x Dd (high Rn: steep and long slopes, high erosion risk).
* Hypsometric curve: plot of relative area vs relative elevation; shows stage of the cycle of erosion (youth, mature, old).
* Slope: average land slope controls velocity of runoff, thus erosion and infiltration.

* Drainage patterns: dendritic (uniform rock), trellis (folded), radial (dome/volcano), rectangular (joints), parallel (steep uniform slope).

> Application:
> Morphometric analysis is used to prioritise sub-watersheds: rank those with highest Dd, Fs, Rn, Rb and lowest Rc, Rf, Re as the most erosion-prone.

2. Soil Erosion

Soil erosion: detachment, transport and deposition of soil particles by agents like water, wind, ice and gravity. Geological (normal) erosion is slow; accelerated erosion caused by human activity is the real problem. India loses about 5334 million tonnes of soil per year (about 16.4 t/ha/yr).

Types of water erosion

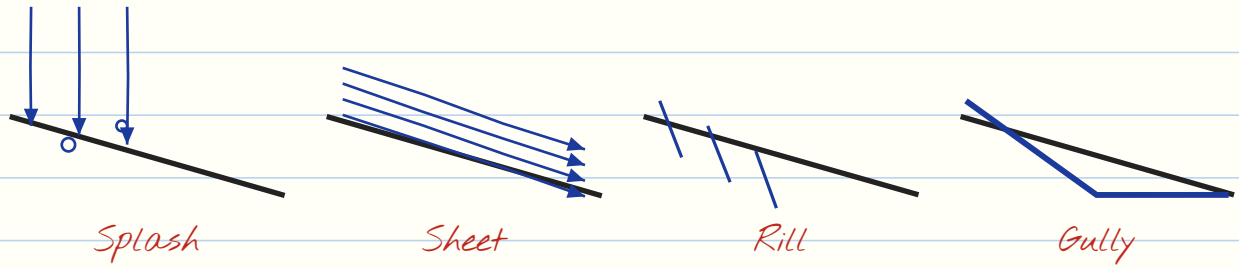


Fig 1: Stages of water erosion

* Splash erosion: raindrop impact detaches and splashes particles (first stage).
* Sheet erosion: thin uniform layer of surface soil removed by shallow runoff; hardly noticed.
* Rill erosion: small finger-like channels, < 30 cm deep, can be removed by tillage.
* Gully erosion: deep channels > 30 cm; cannot be removed by ordinary tillage; classes - small, medium, large (> 9 m).
* Stream-bank erosion, landslides/slip erosion, ravines (Chambal), sea-coast erosion.

* Wind erosion: saltation (50-75 %), surface creep (5-25 %), suspension (3-40 %). Common in arid areas.

Factors affecting erosion

Factor	How it affects erosion
Climate	rainfall amount, intensity, duration and Kinetic energy; wind speed; temperature
Soil	texture, structure, organic matter, permeability; silt and fine sand most erodible, clay and sand less
Topography	slope steepness and length; erosion increases with both
Vegetation	canopy cuts raindrop energy; roots bind soil; litter slows runoff
Land use / management	deforestation, over-grazing, up-and-down slope ploughing, shifting cultivation, mining
Human activity	construction, roads, poor drainage

> Remember:
> Soil loss increases roughly as the 1.5 to 2 power of slope length and slope steepness. Vegetative cover is the single most effective protection.

3. Effects of Erosion on Land Fertility and Capability

On-site effects

* Loss of top soil which is rich in organic matter, N, P, K and micro-nutrients -> fertility falls and yield declines (10-30 %).
* Loss of water-holding capacity and soil depth; root zone becomes shallow.
* Poor structure, crusting, low infiltration -> more runoff, more erosion (vicious circle).
* Gullies cut fields into pieces, reduce cultivable area; rocky and gravelly surface.
* Increased cost of fertiliser to compensate losses; abandonment of land.

Off-site effects

* Siltation of reservoirs, tanks and canals -> reduced storage and life of dams.
* Higher flood peaks, river bed raised, stream channel changes.
* Water pollution by sediment, nutrients and pesticides; eutrophication of lakes.
* Damage to downstream fields by sand casting, harm to aquatic life and fisheries.

Land Capability Classification (USDA, I to VIII)

Class	Suitability	Limitation / practice
I	very good cultivable land	few limits; ordinary practices
II	good cultivable	moderate limits; simple conservation (contour, rotation)
III	moderately good	severe limits, terraces, strip cropping
IV	fairly good, occasional cultivation	very severe limits; special measures
V-VII	not for cultivation: pasture, forest	steep, shallow, wet, stony
VIII	wildlife, recreation, water supply	no agricultural use

Erosion down-grades the capability class: a class II land eroded severely becomes class IV or higher. Land must be used as per capability; sub-classes e = erosion, w = wetness, s = soil, c = climate limitation.

4. Soil Erosion Modelling

Models predict soil loss, plan conservation and prioritise areas. Types: empirical (USLE), conceptual, physically-based (WEPP, EUROSEM, SWAT, AGNPS, ANSWERS).

Universal Soil Loss Equation (Wischmeier and Smith, 1978)

$$A = R\,K\,L\,S\,C\,P \quad (t/ha/yr)$$

* A = computed average annual soil loss (t/ha/yr)
* R = rainfall erosivity factor (MJ mm / ha h yr)
* K = soil erodibility factor (t ha h / ha MJ mm) - for a standard plot
* LS = slope length and steepness factor (dimensionless)
* C = cover and management factor (0 to 1, bare = 1, forest = 0.001)
* P = support (conservation) practice factor (0 to 1, no practice = 1)

Standard unit plot: 22.13 m long, 9 % slope, continuously fallow, tilled up and down slope (K measured on it).

* RUSLE (Revised USLE, 1997): improved R, K, LS, C, P using computers; MUSLE (Williams, 1975) uses runoff volume and peak rate to predict sediment yield per storm.

$$\text{MUSLE: } Y = 11.8\,(Q \times qp)^{0.56}\,K\,LS\,C\,P \quad (Y \text{ in tonnes})$$

* Limitations of USLE: estimates only sheet and rill erosion, average annual basis, no deposition, not for gullies, derived for US conditions.
* Physical models (WEPP) use infiltration, runoff routing and detachment-transport mechanics; need more data but give event and spatial results. GIS + RS used with RUSLE for large watershed maps.

Tolerance:

Soil loss tolerance (T value): maximum permissible soil loss that still allows sustained productivity: 5 to 12 t/ha/yr (India ~ 4.5 - 11.2 t/ha/yr).

5. Erosivity and Erodibility

* Erosivity: ability of rain (or wind) to cause erosion - depends on rain properties. Erosivity of a storm: EI30 = E x I30 where E = total kinetic energy of storm and I30 = maximum 30-minute intensity.

$$KE \text{ per unit rain} = 11.9 + 8.73\,\log10(i) \quad (MJ/ha/mm)$$

i = intensity in mm/h (capped at 76 mm/h). Annual R = sum of EI30 of all storms. In India simple relations like R = 79 + 0.363 P (P = annual rainfall) are used; Fournier index = p^2/P.

* Erodibility: soil's susceptibility to be detached and transported - an inherent property of the soil, not of rainfall.
* Increases with silt and very fine sand content, decreases with organic matter, clay, good structure, high permeability and aggregate stability.
* Nomograph (Wischmeier): K from % silt+very fine sand, % sand, organic matter, structure code and permeability code. Typical K: 0.02 (sandy/clay, low) to 0.07 (silty, high).
* Other indices: dispersion ratio, erosion ratio, clay ratio, aggregate stability, critical shear stress.

6. Sediment Yield and Sedimentation

Sediment yield (SY): total sediment that leaves (outflows from) a watershed measured at the outlet over a period - tonnes/Km2/year. Gross erosion minus deposition on the way.

$$SY = \text{Gross erosion} \times SDR$$

(SDR = sediment delivery ratio)

* SDR < 1: fraction of eroded soil actually reaching the outlet, falls as drainage area increases (more chance of deposition) - typical 0.05 to 0.5.
* Estimation: direct measurement (suspended sediment sampling + bed load), reservoir surveys (capacity loss), rating curve $Qs = a\,Q^b$, empirical formulas, USLE/MUSLE with SDR, models.
* Bed load (rolling, sliding, saltation) about 10-25 % and suspended load the rest, wash load = very fine, from bank/catchment.

Sedimentation in reservoirs

* Deposition of sediment in the reservoir: coarse sediment settles as delta at the head, fine sediment moves forward as turbidity currents and settles near the dam.

* Trap efficiency: percentage of inflowing sediment trapped; Brune curve gives TE as a function of capacity/inflow (C/I) ratio; large reservoirs trap > 90 %.
* Effects: loss of storage (India ~ 0.5 - 1 % per year), reduced hydropower and irrigation, downstream degradation, clogged intakes.
* Control: catchment treatment (soil conservation, check dams, afforestation), silt traps, sediment bypass, flushing / sluicing, dredging, density current venting, designing 'dead storage' for 50-100 years sediment.

Formula:

Design life of reservoir = (Dead storage) / (annual sediment inflow x trap efficiency).

7. Wetlands: Definitions and Role of Water

Wetland: land area where water is the primary factor controlling the environment and plant-animal life. Ramsar Convention (1971): areas of marsh, fen, peatland or water, natural or artificial, permanent or temporary, with water static or flowing, fresh, brackish or salt, including marine water not deeper than 6 m at low tide.

US Fish and Wildlife (Cowardin 1979): lands transitional between terrestrial and aquatic systems where the water table is usually at or near the surface or land is covered by shallow water; must have (a) hydrophytes, (b) hydric soils, (c) wetland hydrology.

Types

* Marsh (herbaceous plants, mineral soils), swamp (woody plants/trees), bog and fen (peatlands - bog gets only rain, fen groundwater too), mangroves, lagoons, estuaries, floodplain wetlands, reservoirs, rice paddies, tanks (human-made).
* India: ~ 4.6 % of area; 85+ Ramsar sites (e.g. Chilika, Keoladeo, Loktak, Sundarbans, Wular).

Role of water in wetland structure and function

* Hydrology is the master variable: depth, duration, frequency and timing of flooding decide soil chemistry (anoxic, reducing), plant species zones and animal communities.
* Water brings nutrients, sediments, seeds and removes wastes; controls oxygen, salinity, pH and temperature.
* Functions: flood storage and peak reduction, groundwater recharge/discharge, water purification (nutrient and sediment retention), shoreline stabilisation, carbon sink, habitat and biodiversity, fisheries, recreation.
* Water fluctuations make pulses of productivity; too little water -> drying and loss, too much -> drowning of vegetation.

8. Wetland Water Budget and Hydroperiod

Water budget: accounting of all water entering, leaving and stored in a wetland in a time period; it is the basis to understand hydrology and manage water level.

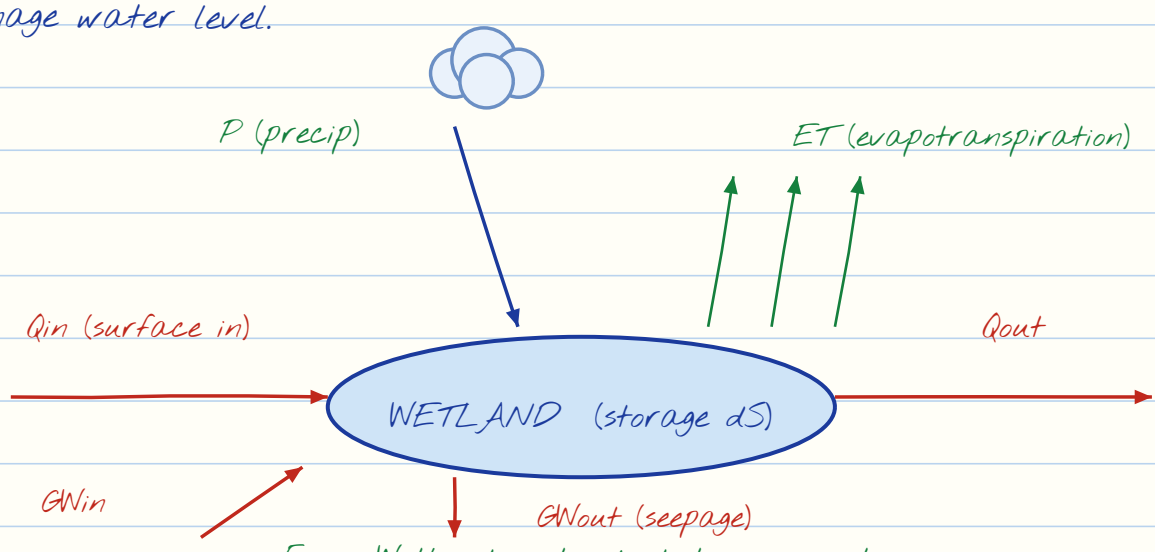


Fig 2: Wetland water budget components

$$dS/dt = P + Qsi + Gi - ET - Qso - Go$$ (storage change)

P = direct precipitation, Qsi/Qso = surface inflow/outflow (streams, overland, tidal), Gi/Go = groundwater inflow/outflow, ET = evapotranspiration, dS = change in storage (surface water + soil water).

Hydroperiod

* Hydroperiod: the seasonal pattern of water-level fluctuation in a wetland - depth, duration, frequency, timing, and rate of rise and fall of water.
* Components that control it: precipitation and evapotranspiration balance, surface water flows, groundwater exchange, tides, basin shape and outlet structure.
* Types: permanently flooded, semi-permanently flooded, seasonally flooded, saturated, temporarily flooded. Hydroperiod decides plant community and habitat for fish, amphibians and birds.
* Flooding duration is often expressed as % of year flooded, wetland criteria - saturated or flooded for >= 5 % of growing season (US) for 1 to 2 weeks.

9. Components of the Water Budget

Inflows, outflows and storage

* Inflows: precipitation, surface inflow (stream, overland sheet flow, tidal flooding), groundwater discharge (springs, seepage), human sources (irrigation return, wastewater).
* Outflows: surface outflow (stream, spillway), evapotranspiration, groundwater recharge (seepage loss), withdrawal.
* Storage: ponded surface water, soil moisture, groundwater, vegetation/biomass. Small change dS shows if wetland is gaining or losing water.

Precipitation and runoff

* Precipitation measured by rain gauges (ordinary and recording), radar, satellite, areal average by Thiessen polygon, arithmetic mean or isohyetal methods.
* Runoff entering a wetland from its catchment: Rational method $Q = C\ i\ A\ /\ 360$ (small area), SCS-CN method for volume, unit hydrograph for

peak and time.

$$\text{SCS-CN: } Q = (P - 0.2S)^2 / (P + 0.8S), \quad S = 25400/CN - 254$$

(mm)

* Curve number CN depends on soil group (A to D), land cover, treatment and antecedent moisture; higher CN = more runoff.

Evapotranspiration (ET)

* ET = evaporation from water/soil surface + transpiration through plants. Often the largest outflow in warm climates; in wetlands ET can equal or exceed Pan evaporation in summer.
* Methods: pan evaporation (ET = Kp x Epan), Penman and Penman-Monteith (energy balance + aerodynamics), Blaney-Criddle (temperature), Thornthwaite, water-budget and eddy covariance, lysimeters.

$$ETc = Kc \times ETo$$

(crop/wetland coefficient)

* ET depends on solar radiation, temperature, humidity, wind speed, vegetation type (cattail, reed), water availability and depth.

> Remember:
> Water budget of a wetland for any period: P + inflows = ET + outflows + dS. Closing the budget gives the unknown term (usually groundwater).

10. Solved Examples (for exams)

Example 1: USLE soil loss

Given: $R = 450$, $K = 0.30$, $LS = 1.2$, $C = 0.25$, $P = 0.5$. Find A.

$$A = 450 \times 0.30 \times 1.2 \times 0.25 \times 0.5 = 20.25 \text{ t/ha/yr}$$

If tolerance $T = 11$ t/ha/yr, then $A > T$ -> conservation needed. Required CP $= T / (R\,K\,LS) = 11 / (450 \times 0.3 \times 1.2) = 0.068$. So $C \times P$ must be reduced from 0.125 to about 0.068 (e.g. better cover + contour bunding).

Example 2: Drainage density

Basin area = 80 sq Km, total stream length = 200 Km, number of streams = 120, perimeter = 42 Km, basin length = 15 Km.

* Dd = 200 / 80 = 2.5 Km per sq Km (fine texture, quick runoff)
* Fs = 120 / 80 = 1.5 per sq Km
* Rf = 80 / 15^2 = 0.356 (elongated basin)
* Rc = 4 x 3.14 x 80 / 42^2 = 0.57

Example 3: Wetland water budget

Over one month: P = 120 mm, ET = 90 mm, surface inflow = 60 mm, outflow = 50 mm, groundwater net = +5 mm (in). Find dS.

$$dS = 120 + 60 + 5 - 90 - 50 = +45 \text{ mm (water level rises)}$$

> Exam tip:
> Always write the unit and the sign of each term. Units of depth (mm) over wetland area are easiest: volume = depth x area.

Quick revision list

* Morphometry: Rb, Dd, Fs, Rf, Rc, Re, Rn - linear, areal, relief.
* Erosion: splash -> sheet -> rill -> gully; USLE A = R K L S C P.
* Sediment yield = erosion x SDR; reservoir life depends on trap efficiency.
* Wetland: hydrology + hydric soil + hydrophytes; budget P + Qin + Gin = ET + Qout + Gout + dS.
* Hydroperiod = seasonal pattern of water depth - the master variable.

UNIT - III

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Syllabus Topics:

Surface water flows: structures and channels | Groundwater - surface water exchange in wetlands | Surface water flows II and wetland hydrology case studies | Flow and mixing in wetlands | Wetland water quality: nutrients, organic/inorganic contaminants, sediments and colloids | Wetland transport models: plug flow, CSTRs and CSTRs in series | Intro to method of moments

1. Surface Water Flows: Structures and Channels

Surface water enters and leaves a wetland as channel flow (streams, ditches), overland sheet flow, tidal flow and flow through hydraulic structures. The way water is controlled decides water level, residence time and hydroperiod.

Open channel flow basics

* Free-surface flow driven by gravity; characterised by depth y, velocity V, discharge Q = A V.
* Continuity: Q = A1 V1 = A2 V2 (steady, incompressible flow).
* Uniform flow: Manning's equation for velocity.

$$V = (1/n)\, R^{(2/3)}\, S^{(1/2)} \qquad \text{(Manning)}$$

n = roughness coefficient (earth channel 0.02–0.03, vegetated wetland 0.05–0.4), R = hydraulic radius = A/P, S = bed slope. Discharge Q = A V.

* Froude number Fr = V / sqrt(g y): Fr < 1 subcritical (tranquil, typical of wetlands), Fr = 1 critical, Fr > 1 supercritical (rapid).
* Reynolds number Re = V R / nu decides laminar (< 500) or turbulent (> 2000) flow. Vegetated wetlands often have low velocities,

'laminar-transitional'.

Flow through vegetation

* Emergent plants (cattail, reed, bulrush) create drag; n is not constant - it falls as depth increases. Stem density and diameter control resistance.
* Drag force on plant: Fd = 0.5 Cd rho a V^2 (a = projected area); Cd about 1 for cylinders.
* Result: slow flow, uniform spreading (sheet flow), more settling and contact with plant surface for water treatment. But flow may short-circuit through open channels (preferential paths).

Hydraulic structures for wetland water control

Structure	Use in wetlands
Weir (sharp, broad crested, V-notch)	control and measure water level/outflow; Q = C L H^1.5 (rectangular), Q = 1.38 H^2.5 for 90 degree V-notch
Flume (Parshall)	flow measurement with small head loss, passes sediment
Culvert	pass flow under roads/embankments; inlet or outlet control
Stop-logs / gates / sluice	adjustable water level; allow drawdown and flushing
Riser pipes / standpipes	fine control of depth in constructed wetlands
Levees, berms, spillways	retain water, protect from floods, emergency overflow
Inlet and outlet manifolds	spread flow evenly across the wetland width
Fish ladders / flap gates	ecological passage, stop tidal back-flow

Formula:

Weir equation (rectangular, free flow): Q = 1.84 L H^(3/2) (SI units, m3/s).

Use contraction correction L = L0 - 0.1 n H for end contractions.

Channels in wetlands

* Natural: creeks, tidal creeks (in salt marsh), meandering sloughs; man-made: ditches, canals, cells linked by channels.
* Channel design: stable, non-eroding velocity (< 0.6 m/s earth); side slope 2:1 to 3:1; free-board; vegetation lining.
* Wetland flow paths and channels (hydroperiod, residence time) are managed by distribution of channels and berms; deep zones and open water zones are used for mixing and re-aeration.

2. Groundwater - Surface Water Exchange in Wetlands

Wetlands are interfaces between surface water and groundwater. They may be recharge, discharge or flow-through (both) types.

Type	Flow direction	Remarks
Discharge wetland	groundwater -> wetland	water table higher than wetland; springs, fens, stable water level and chemistry
Recharge wetland	wetland -> groundwater	water table below the wetland bottom; seasonal; depressions (prairie potholes)
Flow-through	in on one side, out on the other	both; typical for lakes and riparian wetlands

Darcy's law for seepage

$$Q = K A (dh/dl) \quad \text{(Darcy)}$$

K = hydraulic conductivity of the bed material (clay 1e-9 m/s; peat 1e-6; sand 1e-4), A = seepage area, dh/dl = hydraulic gradient between wetland stage and adjacent water table.

* Seepage rate per unit area: $q = K i$. Sign gives gain (+) or loss (-) in budget. Often a small term, but important in dry season and for water quality (groundwater is stable temperature and rich in minerals).
* Measurement: piezometers/mini-piezometers and nested wells (head difference), seepage meters (direct flux), temperature profiles, tracers

(Rn-222, chloride), water balance residual, numerical models (MODFLOW).

* Factors: water table slope, bed sediment (clay layer, peat), vegetation ET (daily drawdown of the water table), seasonal rainfall, pumping nearby, tides.
* Hyporheic exchange: mixing of surface water with shallow subsurface water through bed sediments - improves nutrient retention (denitrification).
* Management: do not drain or pump groundwater near wetland, maintain recharge areas in the catchment, clay liners in constructed wetlands to avoid leakage.

3. Surface Water Flows II and Wetland Hydrology Case Studies

* Sheet flow in the Everglades (Florida): very flat slope 3 cm/Km, vegetation (sawgrass) drag controls flow; compartmentalisation by canals and levees has changed the hydroperiod; Comprehensive Everglades Restoration Plan restores flow volume, timing and distribution.
* Chilika lake (Odisha, India): brackish lagoon, Ramsar site; inlet to sea was choked by silt, salinity and fish catch fell; hydraulic intervention opened new mouth (2000) -> restored salinity regime, water exchange and fisheries; weed decreased. Example of hydrology-based restoration.
* East Kolkata Wetlands (Ramsar): natural sewage-fed fishery; waste water flows through channels to ponds - low-cost treatment of about 750 million litres per day; fish and vegetables grown.
* Keoladeo Ghana National Park (Bharatpur): artificial wetland whose water is controlled by sluice gates and canals from Ajan Bund; decline in water due to diversions caused loss of bird habitat, management of inflow is vital.
* Loktak lake (Manipur): Ithai barrage on Manipur river changed hydroperiod, killed phumdis (floating biomass), submerged farms; shows effect of a structure on a wetland.

* Constructed treatment wetlands (Arcata, California; Kolkata; Coimbatore): flow, depth and residence time are designed for removing BOD, N and P.

Case study lessons:

Lessons: restore natural hydroperiod, maintain connectivity with river/sea, avoid over-regulation, monitor water level and quality, involve local community.

4. Flow and Mixing in Wetlands

Mixing decides how fast pollutant spreads and how long water stays inside the wetland (residence time), thus treatment performance.

Nominal residence time $t = V / Q$ (volume / flow rate)

* Actual residence time differs due to dead zones, short-circuiting and variations in depth: hydraulic efficiency = t_peak / t_nominal (e_v = t_mean / t_nominal).

Mixing processes

* Advection: transport by mean flow (velocity V).
* Molecular diffusion: very slow random motion, coefficient D about 1e-9 m2/s.
* Turbulent diffusion and dispersion: caused by eddies, velocity differences, wind, plants - much larger than molecular diffusion; longitudinal dispersion coefficient E.
* Vegetation increases mixing (wake of stems) but reduces velocity; open water zones have wind mixing and short-circuiting.

Advection-dispersion: $dC/dt + V\, dC/dx = E\, d2C/dx2 - KC$

C = concentration, K = first-order decay (reaction) constant. Peclet number Pe = V L / E tells which process dominates: Pe large -> plug flow like; Pe small -> well mixed.

Tracer studies

* Inject a conservative tracer (Rhodamine WT, bromide, lithium chloride, fluorescein) at the inlet and measure concentration versus time at the outlet -> RTD curve (residence time distribution), E(t).
* Mean residence time = first moment of RTD; variance = second central moment gives the spread (dispersion).
* Common patterns: early peak = short-circuit, long tail = dead zones; ideal curve = narrow peak at t = V/Q.

5. Wetland Water Quality: Contaminants

Nutrients (N and P)

* Nitrogen forms: organic N, ammonium NH_4^+, nitrite, nitrate NO_3^-, N_2 gas. Removal by plant uptake, nitrification (aerobic) then denitrification (anoxic, NO_3^- -> N_2), ammonia volatilisation, sedimentation of organic N.
* Phosphorus: dissolved ortho-P, particulate P. Removal by adsorption to soil (Fe, Al, Ca), precipitation, sedimentation, plant uptake and peat accretion. Long-term sink is limited; released again under anoxia.
* Excess N and P -> eutrophication, algal blooms, low DO.

Organic and inorganic contaminants

* Organic: BOD, COD, pesticides (DDT, atrazine), PAHs, PCBs, pharmaceuticals, oil - removed by sedimentation, microbial degradation (biofilm on plants), adsorption, photolysis and volatilisation.
* Inorganic: heavy metals (Pb, Cd, Cu, Zn, Hg, As), salts, acid mine drainage - removed by precipitation as sulphides/hydroxides, adsorption on organic matter and Fe/Mn oxides, plant uptake (phytoremediation, e.g. water hyacinth, Typha).
* Pathogens: removed by sedimentation, UV, predation, natural die-off.

Sediments and colloids

* **Sediments:** suspended solids (clay, silt) settle when velocity falls in vegetation; Stokes law for settling velocity.

$$V_s = g(\rho_s - \rho_w) d^2 / (18 \mu) \quad \text{(Stokes)}$$

* **Colloids:** particles 1 nm – 1 micron (clay, humic substances, bacteria, iron oxyhydroxides). Stay in suspension due to surface charge, carry sorbed metals, P and organics (facilitated transport), removed by coagulation/flocculation, filtration on plants and biofilm.
* Sediment burial reduces the loading but eventually deposited sediment fills the wetland and reduces storage volume and capacity.

Pollutant	Main removal in wetland
TSS	settling, filtration by plants
BOD	microbial degradation on biofilm
Nitrogen	nitrification-denitrification, uptake
Phosphorus	sorption, sedimentation, accretion
Metals	precipitation, adsorption, uptake
Pathogens	settling, UV, die-off

6. Wetland Transport Models I

Reactor models represent the wetland as an idealised reactor: the first-order decay of pollutant with rate constant K is used for design.

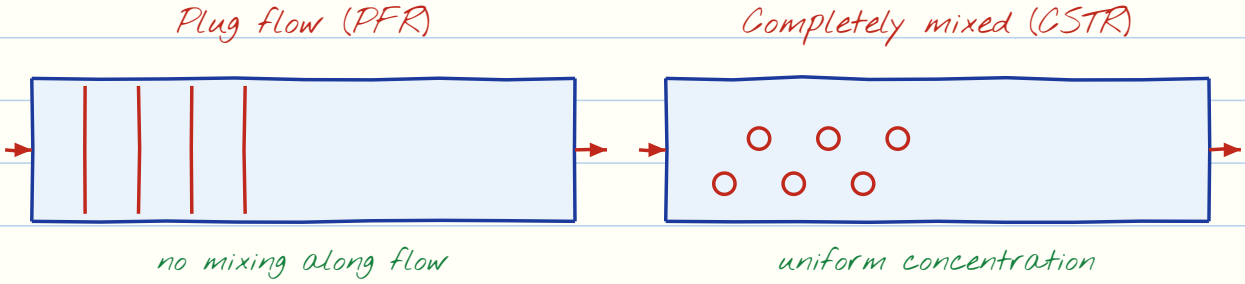


Fig 1: Two ideal reactors

A. Plug flow reactor (PFR)

* No mixing in the direction of flow; each slice of water travels as a plug; all fluid stays exactly $t = V/Q$.

$$C_{out} = C_{in} \exp(-K t)$$ (first-order)

This is the most efficient reactor: lowest outlet concentration for a given K and t. Used for long narrow wetland cells with length : width > 3:1.

B. Continuous stirred tank reactor (CSTR)

* Perfect mixing; outlet concentration = concentration inside the tank everywhere.
* Mass balance (steady): $Q C_{in} = Q C + K C V \rightarrow$

$$C_{out} / C_{in} = 1 / (1 + K t)$$

* Less efficient than PFR for the same t (because incoming clean/dirty water is instantly diluted down to the lower level where the reaction rate is low).
* Unsteady: $V dC/dt = Q (C_{in} - C) - KVC$; solution for step input $C = C_{final} (1 - \exp(-t/\tau))$ with $\tau = V/Q$.

C. CSTRs in series (tanks in series model)



$C_n / C_0 = 1 / (1 + t/N)^N$ (N tanks in series)

Fig 2: N tanks in series

$$C_N / C_0 = 1 / (1 + K t / N)^N$$

* Each tank has volume V/N; N = 1 gives a single CSTR, N -> infinity gives a PFR. So real wetlands (partial mixing) lie between: use N from tracer data.

* Relation with dispersion: variance of RTD sigma^2 = t^2 / N, so N = t^2 / sigma^2. For Pe: N = Pe / 2 (approx).
* Kadlec and Wallace tanks-in-series (TIS) model with first order area-based K: C* = (C - C*) ... P-K-C* model, includes background concentration C*.

$$(C_{out} - C^*) / (C_{in} - C^*) = (1 + K / (N q))^{(-N)} \quad \text{(P-K-C*)}$$

q = hydraulic loading rate = Q/A (m/yr), K = areal rate constant (m/yr), C* = irreducible background concentration.

> Compare:
> Efficiency order for same K and t: PFR > CSTRs in series > single CSTR. Series arrangement of cells improves performance.

7. Introduction to Method of Moments

Moments summarise a tracer response curve C(t) (or RTD) by a few numbers instead of fitting a full curve.

$$M_n = \int t^n C(t)\, dt \quad \text{(n-th absolute moment, from 0 to infinity)}$$

* Zeroth moment M0 = integral C dt = area under curve -> mass recovery (M0 Q = mass injected if conservative).
* First moment (mean) : mean residence time tm = M1 / M0 (centroid of curve).
* Second central moment (variance): sigma^2 = M2/M0 - tm^2 -> spread of the curve (dispersion).
* Third moment (skewness): asymmetry; long tail indicates dead zones or storage.

$$\text{Effective volume } V_{eff} = Q\, tm \quad ; \quad \text{dead volume fraction} = 1 - V_{eff} / V$$

* Dispersion number: sigma^2 / tm^2 = 2 (E/VL) - 2 (E/VL)^2 (1 - exp(-VL/E)) for closed vessel; N tanks: sigma^2/tm^2 = 1/N.
* Procedure: (1) inject pulse tracer, (2) sample outlet at time intervals, (3) compute C vs t, (4) numerical integration by trapezoid rule, (5)

calculate M0, M1, M2, (6) find tm, sigma^2, N, Pe.

Mini example

tm = 10 days and sigma^2 = 25 days^2 -> N = tm^2 / sigma^2 = 100 / 25 = 4 tanks. With nominal t = 12 d, effective volume ratio = 10/12 = 0.83 -> 17 % dead zone.

> Take-away:
> Method of moments is a quick way to get residence time, dispersion and number of tanks without fitting complex models.

8. Design Relations for Treatment Wetlands

* Free water surface (FWS) wetland: shallow water 0.1 - 0.6 m, emergent vegetation; area A = Q / q.
* Subsurface flow (SSF) wetland: water flows through gravel/sand bed under the surface (horizontal or vertical), less mosquito problem, smaller area but costlier media.

Item	FWS wetland	SSF wetland
Water depth	0.1 - 0.6 m open water	0.3 - 0.6 m below gravel surface
Residence time	5 - 14 days	3 - 7 days
HLR	0.01 - 0.05 m/d	0.02 - 0.1 m/d
Pros	cheap, habitat value, large volume	odour free, thermal protection, compact
Cons	mosquitoes, more area	clogging, high media cost

* Area required (first-order plug flow): A = Q ln(Cin / Cout) / (K y n) with K in per day, y = depth, n = porosity.

$$A = Q \ln(C_{in}/C_{out}) / (K_A) \quad (K_A = \text{areal rate constant, m/d})$$

* Typical aspect ratio L : W = 3:1 to 10:1 to avoid short circuiting, baffles, islands and deep zones improve mixing.

* Water depth, bed slope (0.5 – 1 %), inlet distribution, vegetation density (> 60 % cover) are the design factors, also freeboard for storm flow.

9. Solved Examples and Exam Hints

Example 1: Residence time

A wetland of area 2 ha and average depth 0.4 m with porosity 0.9 receives 800 m3/day. Find t.

$$V = 20000 \times 0.4 \times 0.9 = 7200\ m^3\ ;\ t = V/Q = 7200/800 = 9\ days$$

Example 2: Pollutant removal (PFR vs CSTR)

Cin = 40 mg/L, K = 0.3 /day, t = 9 days.

* PFR: Cout = 40 exp(−0.3 × 9) = 40 × 0.067 = 2.7 mg/L
* CSTR: Cout = 40 / (1 + 0.3 × 9) = 40 / 3.7 = 10.8 mg/L
* 3 CSTRs in series: Cout = 40 / (1 + 0.3 × 9 / 3)^3 = 40 / (1.9)^3 = 40 / 6.86 = 5.8 mg/L

So the performance order PFR (2.7) < TIS (5.8) < CSTR (10.8): a lower concentration = better.

Example 3: Manning flow

Wide vegetated channel: n = 0.2, R = 0.3 m, S = 0.0002.

$$V = (1/0.2) \times 0.3^{(0.667)} \times 0.0002^{(0.5)} = 5 \times 0.448 \times 0.01414 = 0.032\ m/s$$

Example 4: Darcy seepage

K = 1e-6 m/s, head difference 0.3 m over 3 m path, area 1000 m2: Q = 1e-6 × 1000 × 0.1 = 1e-4 m3/s = 8.6 m3/day (recharge if wetland head higher).

Likely questions

* Explain groundwater–surface water exchange in wetlands (draw discharge, recharge, flow-through types).
* Derive the CSTR and tanks-in-series equations and compare with PFR.
* Explain method of moments for a tracer test with an example.
* Discuss nutrient and contaminant removal mechanisms in wetlands.
* Case study: Chilika / Everglades / East Kolkata wetlands.

10. Quick Revision Chart

Model	Mixing	Equation (first order)	Remark
PFR	none along flow	Cout/Cin = exp(-Kt)	best efficiency
CSTR	complete	Cout/Cin = 1/(1+Kt)	lowest efficiency
TIS (N tanks)	partial	1/(1+Kt/N)^N	N = t^2 / sigma^2
Dispersion (PFD)	axial dispersion E	Peclet number Pe = VL/E	Pe = 2N approx

Water quality monitoring in wetlands

* Sampling points: inlet, intermediate, outlet, and background (reference); grab and composite samples; flow-weighted for loads.
* Parameters: field (pH, DO, temperature, EC, turbidity), lab (BOD, COD, TSS, TN, TP, NH4-N, NO3-N, coliforms, metals).
* Load = concentration x flow: L = C Q (Kg/d); removal efficiency % = (Lin - Lout) / Lin x 100; areal removal rate = (Lin - Lout) / A.
* Seasonal variation: warm season gives faster microbial removal; winter plant die-back releases nutrients; storms dilute and flush.
* Long-term issues: accumulation of sediments and metals, clogging in SSF beds, plant harvesting, mosquito and odour control.

Last line:

To remember transport models: plug = no mixing (best), CSTR = full mixing (worst), series of CSTRs = in between; method of moments gives tm, variance and N from tracer data.

UNIT - IV

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Syllabus Topics:

Wetland hydrologic assessment: physical and biological processes | Anthropogenic and climate change impacts on wetland hydrology | Modeling wetland hydrology, hydraulics and hydrodynamics | Introduction to wetland treatment systems design | Water harvesting: rainwater, catchment harvesting, harvesting structures | Model watershed | Government and NGO projects

1. Wetland Hydrologic Assessment

Hydrologic assessment: collecting and analysing data on water sources, flows, levels and storage to understand how a wetland functions and how it responds to change. It is the first step in wetland conservation, restoration and design.

Steps of assessment

1. Delineate the wetland and its contributing catchment (topographic map, DEM, GIS).
2. Collect climate data: rainfall, temperature, solar radiation, wind, humidity -> ET.
3. Measure surface inflow and outflow (gauges, weirs, flumes, current meter), water levels (staff gauge, data loggers), tidal records.
4. Measure groundwater: observation wells, piezometers, seepage meters.
5. Prepare water budget, compute hydroperiod and residence time, check closure and uncertainty.
6. Link hydrology to soils (redox features, mottles), vegetation (indicator species) and fauna.

7. Use remote sensing (NDWI, SAR, Landsat/Sentinel) to map extent and seasonal change of water spread.

Physical processes

* Precipitation, runoff, infiltration, ET, groundwater exchange, tidal and wind effects, sedimentation and erosion, thermal stratification, evaporation losses.
* Sediment deposition changes depth and storage; erosion of banks changes channels; flow patterns change with vegetation drag.
* Wind-induced seiche and mixing in open-water zones; ice cover in cold regions.

Biological processes

* Vegetation: transpiration and shading modify ET and temperature, plant growth changes roughness (flow resistance) and traps sediment, peat accumulation raises the bed over time.
* Microbes: decomposition, nitrification-denitrification, sulphate reduction -> control water chemistry and carbon storage.
* Animals: beaver dams, crayfish burrows, fish and birds change hydrology and nutrient movement, invasive plants like water hyacinth increase ET and block flow.

> Eco-hydrology:
> Hydrology (physical) and ecology (biological) are coupled: hydroperiod shapes biology and biology feeds back on hydrology (eco-hydrology).

2. Anthropogenic and Climate Change Impacts

Human (anthropogenic) impacts

Activity	Hydrological impact
Drainage and reclamation	lower water table, loss of wetland area, reduced storage and flood buffer
Dams, barrages, diversions	altered flow regime, loss of floods/pulses, salinity change (Loktak, Keoladeo)
Urbanisation	impervious surface -> high runoff, polluted storm water, shrinking wetlands
Agriculture	nutrient/pesticide load, irrigation withdrawals, drainage ditches
Groundwater over-pumping	reduces seepage to wetland, springs dry
Deforestation in catchment	more erosion and siltation, flashy flows
Sewage and industry	pollution, eutrophication, loss of DO
Invasive species / over-fishing	choke channels, change ET and food web

Climate change impacts

* Temperature rise: higher ET, more evaporation loss, shorter hydroperiod, warmer water -> lower DO, algal blooms.
* Rainfall changes: more extreme storms and longer dry spells, shifts in monsoon timing, more floods and droughts.
* Sea-level rise: salt-water intrusion in coastal wetlands, loss of mangroves/marshes unless they can migrate inland.
* Glacier and snow-melt change affects Himalayan wetlands (Wular, Pangong, Tso Moriri).
* Feedback: wetlands emit methane (CH_4) and store carbon (peat, blue carbon), drying releases CO_2.
* Adaptation: restore wetlands as natural flood buffers, maintain environmental flows, protect buffer zones, integrated river-basin management, monitoring.

> Note:
> Impacts are often cumulative: human use + climate change together push wetlands over thresholds, so management must be adaptive.

3. Modeling Wetland Hydrology, Hydraulics, Hydrodynamics

Models are mathematical tools to simulate water level, flow and transport in a wetland for design, forecasting and impact assessment.

A. Hydrologic models (water budget)

* Lumped water-balance model: $dS/dt = P + Qin - ET - Qout +/- G$. Solve with time step (daily). Needs rainfall, ET, stage-storage and stage-discharge curves.
* Rainfall-runoff in catchment: SCS-CN, unit hydrograph, HEC-HMS, SWAT, VIC.
* Stage-storage curve S(h) from bathymetry/DEM; outflow rating curve Q(h) of weir/culvert.

$$S(t+dt) = S(t) + (P + Qin - ET - Qout)\, dt$$

This reservoir-routing (level-pool or Modified Puls) approach gives the stage hydrograph.

B. Hydraulic models (flow in channels and structures)

* 1-D Saint-Venant equations (continuity + momentum) for unsteady open-channel flow:

$$dA/dt + dQ/dx = q_lat \; ; \quad dQ/dt + d(Q^2/A)/dx + gA\, dh/dx + gA\, Sf = 0$$

* Tools: HEC-RAS (1-D, 2-D), MIKE 11, SWMM; steady gradually varied flow by standard step method.
* Vegetated flow: roughness from stem density, drag coefficient, n depends on depth.

C. Hydrodynamic models (2-D / 3-D)

* Depth-averaged shallow water equations in 2-D (wetland with wide shallow areas); 3-D for stratified lakes and estuarine tidal marshes.
* Tools: MIKE 21, Delft3D, TELEMAC, EFDC, SFWMM (Everglades), MODFLOW (coupled groundwater), WASP and CE-QUAL for water quality; WETLAND-DNDC for biogeochemistry.
* Process: (1) conceptual model, (2) choose software, (3) grid and boundary conditions (inflow, tide, wind), (4) calibration (adjust n, K), (5) validation with independent data, (6) sensitivity and uncertainty, (7) scenario simulation (climate, management).
* Performance statistics: Nash-Sutcliffe efficiency NSE, RMSE, R^2, percent bias.

> Selecting a model:
> Model choice depends on purpose and data: lumped budget for water level, 1-D for channel, 2-D for sheet flow and mixing, coupled models for ground-surface exchange.

4. Introduction to Wetland Treatment Systems Design

Constructed (treatment) wetlands are engineered systems that use natural processes of wetland vegetation, soils and microbes to treat waste water, storm water, agricultural drainage, mine drainage.

Types

* Surface flow (FWS) wetlands: open water with emergent plants.
* Horizontal subsurface flow (HSSF): water below gravel bed.
* Vertical flow (VF): intermittent dosing from top, good aeration, nitrification.
* Hybrid systems (VF + HSSF) for N removal; floating treatment wetlands, reed beds for sludge drying.

Design steps

1. Define objective: pollutant, inflow Q, influent and target effluent quality (Cin, Cout).
2. Pre-treatment: screens, septic tank/settler to remove solids and grease.
3. Select type, site (flat, away from houses, soil permeability), climate.
4. Size the area: $A = Q \ln(Cin/Cout) / K$ (plug flow) or P-K-C* model for N tanks.
5. Depth (0.3 - 0.6 m), L: W ratio, bed slope, number of cells in series/parallel, inlet and outlet structures.
6. Media: gravel size 10 - 20 mm for SSF; liner (clay or HDPE) to protect groundwater.
7. Vegetation: Typha (cattail), Phragmites (reed), Scirpus, Canna, Colocasia, Vetiver; plant at 4 - 9 per sq m.
8. O and M: harvesting, sludge removal, water level control, mosquito control, monitoring.

Parameter	Typical design value
Hydraulic loading rate	0.02 - 0.1 m/day (SSF), 0.01 - 0.05 (FWS)
Organic loading	< 80 Kg BOD/ha/day (FWS), < 100 (SSF)
Area per person (domestic)	2 - 5 sq m (SSF) ; 5 - 10 (FWS)
Retention time	3 - 14 days
Removal BOD / TSS / N / P	70-90 % / 70-90 % / 30-60 % / 30-60 %

Mechanisms:

Treatment wetland mechanisms: sedimentation, filtration, adsorption, microbial degradation, plant uptake, nitrification-denitrification, UV and die-off.

5. Water Harvesting

Water harvesting: collection, storage and use of rain water (runoff) from rooftops, land surface or rock catchments for beneficial use or to recharge groundwater - 'catch the rain where it falls'.

A. Rainwater harvesting (RWH)

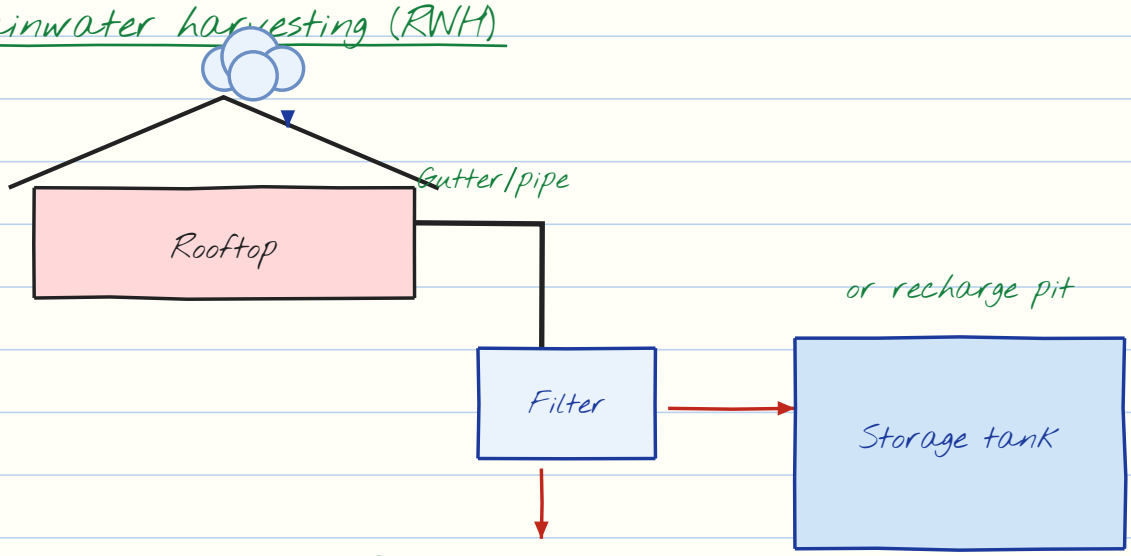


Fig 1: Rooftop rain water harvesting

* Components: catchment (roof), gutters and down pipes, first-flush diverter (discards the first 1-2 mm of rain), filter (sand, gravel, charcoal), storage tank (ferro-cement, RCC, plastic) or recharge structure.
* Recharge structures: recharge pit, trench, recharge well/borewell, injection well, percolation tank.

$$\text{Harvest } V = \text{Rainfall (m)} \times \text{Roof area (m2)} \times \text{runoff coefficient}$$

(C = 0.8 roof)

Example: 100 sq m roof, 800 mm rain, C = 0.8 -> V = 0.8 x 100 x 0.8 = 64 m3 = 64,000 litres per year.

* Tank size = daily demand x dry period days, maintain water quality by cleaning and chlorination.
* Advantages: free source, reduces storm water, recharges groundwater, improves quality (low TDS), cheap and decentralised. Mandatory for buildings in many cities (Chennai, Bengaluru, Hyderabad).

B. Catchment (runoff) harvesting

* Collect runoff from natural/treated land catchments into storage: ponds, tanks, small reservoirs.
* Catchment treatment: compaction, clearing, plastic/butyl lining, bitumen, concrete, chemical (sodium carbonate) to raise runoff coefficient, Microcatchment (negarim), contour bunds, contour furrows for trees.
* Pond capacity: V = (catchment area x rainfall x runoff coeff.) - losses; design to hold 1-2 irrigations.
* Types by scale: micro-catchment (in situ), macro-catchment (50 - 1000 sq m catchment for one field), floodwater harvesting (spate irrigation), roof and ground catchment.

6. Harvesting Structures

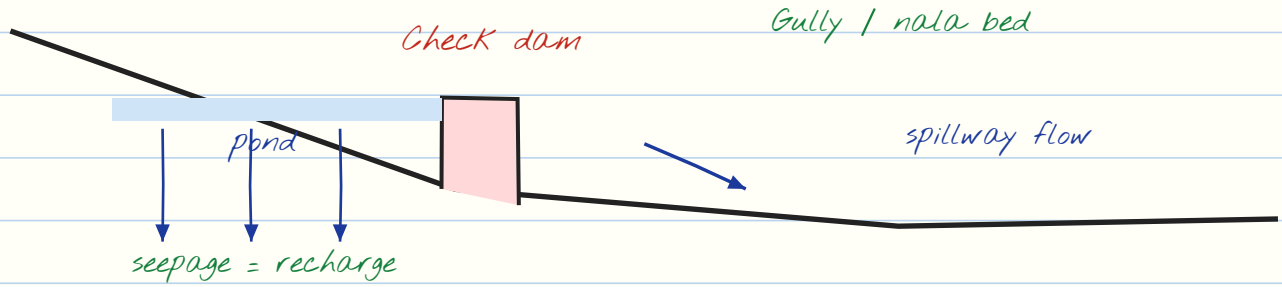


Fig 2: Check dam with pond and recharge

Structure	Features / use
Farm pond	dugout or embankment 0.5–3 m depth; protective irrigation, fish; lined to cut seepage
Contour bund / trench	on slopes 2–6 %; hold runoff, increase infiltration; spacing by slope
Gully plug / loose boulder	small structures across gullies; trap silt, use local stone
Check dam (masonry/earthen/gabion)	across nala; 0.5–3 m high; pond + recharge; spacing by slope
Nala bund	earthen bund across a nala with spillway; 1 – 3 m; for streams with 3 – 10 % slope
Percolation tank	pond on permeable ground for groundwater recharge; downstream wells benefit
Sub-surface dyke (underground bandhara)	impervious wall below river bed to stop sub-surface flow in sandy beds
Khadin / johad / tanka	traditional Rajasthan structures; Khadin = bunded field, johad = earthen check dam, tanka = underground tank
Stop dam / weir, Kolhapur-type (KT) weir	barrage with gates to store river water

* Site selection: narrow section with good foundation, large catchment, stable banks, permeable strata downstream (for recharge), away from habitat.
* Design of check dam: height H, length L of spillway ($Q = 1.7\ L\ H^{1.5}$), apron and cut-off wall, wings, protection of toe from scour. Spacing: Vertical interval = 1 – 2 m, Spacing = H / slope.
* Maintenance: de-silting, repairs of damaged section, checking of leakage, community ownership and water user group.

Sequence:

Ridge-to-valley sequence: contour trenches on hills -> gully plugs -> check dams in drainage lines -> farm ponds and percolation tanks in lower reach.

7. Model Watershed

Model watershed: a demonstration watershed developed in a participatory manner with the best technology so that other villages can learn and replicate. It shows integrated soil, water, crops, livestock and institutions.

* Features: fully treated ridge to valley, community institutions (watershed committee, SHGs, user groups), self-sustained maintenance fund, monitoring of runoff and sediment at the outlet, crop diversification and income generation.
* Examples in India: Ralegan Siddhi (Maharashtra, Anna Hazare) - water conservation + social reforms; Hiware Bazar (Maharashtra, Popatrao Pawar) - 'model water-rich village'; Sukhomajri (Haryana) - silt control + community sharing of water; Ralegaon; Adgaon; Kothapally (Telangana - ICRISAT consortium); Rajasthan - Tarun Bharat Sangh (Rajendra Singh) revival of Arvari river by johads; Jhabua (MP) - Livelihood-based approach.
* ICAR model watersheds in 47 centres; NRAA and CRIDA (Hyderabad) developed best practice; 'Watershed plus' approach.
* Success factors: leadership, participation, transparency, equity, gram sabha decision, social fencing (ban on grazing, ban on tree felling, ban on water-hungry crops), simple technology, capacity building, linkage to markets.

Village	Key actions	Result
Ralegan Siddhi	check dams, percolation tanks, afforestation, grazing ban, prohibition	water table up, double crops, migration stopped
Hiware Bazar	trenches, 'water budget' per year, crop planning	poverty fell, income per capita rose many times
Sukhomajri	Sukhna lake silt control by dams + water rights	sediment in lake reduced > 90 %
Arvari (Alwar)	300+ johads, community Arvari Sansad	river flows perennially again

8. Government and NGO Projects

Government programmes

* DPAP (Drought Prone Areas Programme, 1973), DDP (Desert Development Programme, 1977), IWDP (Integrated Wastelands Development Programme, 1989).
* NWDPRA (National Watershed Development Project for Rainfed Areas, 1990-91), River Valley Project (RVP) and Flood Prone River (FPR) schemes for catchments of dams.
* Hariyali guidelines (2003): Panchayati Raj institutions as the planning and implementing agency.
* IWMP (2009-10) under Dept of Land Resources, then PMKSY - WDC (2015-16): Pradhan Mantri Krishi Sinchayee Yojana (Har Khet Ko Pani, 'more crop per drop'), WDC-PMKSY 2.0 (2021-26).
* MGNREGA - works on water conservation, water harvesting and drought-proofing; Jal Shakti Abhiyan - Catch the Rain; Atal Bhujal Yojana (community groundwater management in 7 states); Mission Amrit Sarovar (75 water bodies per district); National Mission for Clean Ganga, Namami Gange; National Wetland Conservation Programme (NPCA).

* Common Guidelines (2008): shared cost 90:10 (Centre:State), project cost Rs 12,000 - 15,000 per ha; 1 % M and E, 10 % administration, 5 % institution, 50 % works.

NGO and international projects

* BAIF (Pune), WOTR (Pune), Tarun Bharat Sangh (Rajasthan), Foundation for Ecological Security (FES), Sadguru Foundation (Gujarat), MYRADA (Karnataka), Aga Khan Rural Support Programme, DHAN Foundation (tanks), PRADAN, Watershed Support Services and Activities Network (WASSAN).
* Donor assisted: Indo-German Watershed Programme (Maharashtra), Indo-Swiss, World Bank-assisted Karnataka Watershed Development (Sujala), Andhra Pradesh Rural Livelihood Project, DFID, IFAD, ICRISAT 'Consortium approach', NABARD (WDF - Watershed Development Fund).
* Role of NGOs: facilitation, social mobilisation, training, technical support, monitoring, building community institutions, linking with government schemes.

Success:

Common elements of successful projects: people's participation, ridge-to-valley treatment, equity, capacity building, convergence of schemes, sustainability fund.

UNIT - V

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Rain water management | Planning and operation of irrigation systems | Conjunctive use of water | Participatory irrigation management and WRM | Water management policy during droughts | Predicting effect of water shortage on crops | Water footprint of crops and its applications | Blue, green and grey water footprint

1. Rain Water Management

Rain water management: planned handling of rain (the primary source of water) so that maximum useful water is stored in the soil, ponds and aquifers and the excess is drained safely. About 55 % of India's net sown area is rainfed, so it is the Key to food security.

Components

* In situ conservation: store rain in the soil profile - contour farming, bunds, broad bed and furrow, mulching, tillage, tied ridges, land levelling, deep ploughing.
* Ex situ harvesting: runoff stored in farm ponds, check dams, percolation tanks and used for supplemental irrigation.
* Recharge: rooftop and recharge pits, trenches, injection wells; Managed Aquifer Recharge (MAR).
* Safe disposal: grassed waterways, diversion drains, drop structures, check dams to carry excess flow without erosion.
* Crop management: suitable crops and varieties, sowing at onset of monsoon, mid-season correction, mulching, intercropping, weed control, anti-transpirants.

Effective rainfall = Rainfall - Runoff - Deep percolation - Interception loss

* Effective rainfall methods: USDA-SCS (monthly), FAO/AGLW formula: Pe = 0.6 P - 10 (P < 70 mm/month); Pe = 0.8 P - 24 (P > 70 mm/month).
* Dependable rainfall: 75 % probability rainfall (rain expected 3 of 4 years) used in planning (Weibull plotting position m / (n+1)).
* Water Use Efficiency WUE = yield (Kg/ha) / water used (mm or m3). Rain-water use efficiency RWUE = yield / rainfall.

Remember:

Rain water management = capture + store + use efficiently + recharge.

Practice: Conserve in the soil first, then store extra runoff in ponds for protective irrigation.

2. Planning and Operation of Irrigation Systems

Irrigation: artificial application of water to soil to supply moisture essential for plant growth. Planning gives the right project, operation gives right use.

Planning of an irrigation system

1. Water availability: surface (river, reservoir, tank), groundwater, 75 % dependable yield, losses, other uses; reservoir simulation.
2. Command area and soils: Gross Command Area (GCA), Culturable Command Area (CCA) = GCA - unculturable; Net sown area; land capability and drainage.
3. Crop water requirements: crop pattern and calendar, ETO and Kc, effective rainfall, losses, leaching.
4. Selection of system: gravity canal, lift, tank, well, drip, sprinkler; storage + diversion headworks.
5. Design of canal network: main canal, branches, distributaries, minors, water courses and field channels; capacity from peak demand (duty).
6. Economic analysis: benefit-cost ratio, IRR, financial viability, social and environmental impact.

Key terms and formulas

$$ETc = Kc \times ETO \; ; \quad NIR = ETc - Pe \; ; \quad GIR = NIR / Ea$$

(Ea = field application efficiency)

* Duty (D): area irrigated per unit discharge (ha per cumec), Delta (Delta): total depth of water required by crop in base period B (days).

$$Delta = 8.64 \; B / D$$

(Delta in m, B in days, D in ha/cumec)

* Base period (B), Kor depth/period, Kor watering for first irrigation, Consumptive use Cu, Irrigation interval = (Field capacity - Allowable depletion) / ETc.
* Efficiencies: conveyance Ec = water delivered to field / water at head; application Ea = water stored in root zone / water delivered; project efficiency = Ec x Ea (typical canal 30-40 %, drip 85-90 %).

Method	Efficiency	Suited for
Surface (border, furrow, basin)	40 - 60 %	level fields, cereals, rice
Sprinkler	70 - 80 %	undulating land, pulses, vegetables
Drip / micro	85 - 95 %	orchards, row crops, water scarce area
Sub-surface / pitcher	90 %	arid, small gardens

3. Operation of Irrigation Systems

* Water delivery methods: continuous flow, rotational (Warabandi - North India, fixed turn by time in proportion to land holding), demand supply, arranged delivery.
* Warabandi: fixed weekly roster - each farmer gets water turn (hours = area x time per ha), ensures equity at tail end.
* Canal operation: flow regulation at head regulators and cross regulators, scheduling of canal closures, monitoring discharge with weirs/flumes,

SCADA/automation, volumetric supply.

* Reservoir operation: rule curve - release as per demand and flood storage, conjunctive operation with wells.
* Maintenance: de-silting, repair of lining, weed removal, dividing structures, drainage clearing, Operation and Maintenance (O and M) funded by water charges.
* Problems: tail-end deprivation, seepage and waterlogging, salinisation, low water-use efficiency, unauthorised drawing, poor O and M funds.
* Modernisation: lining, pressurised pipe network, Pressure irrigation, on-line monitoring, remote sensing for crop area and ET, SCADA, farmers' water-user associations.
* Irrigation scheduling: when and how much - based on soil moisture (tensiometer, gypsum block), climatic approach (IW/CPE ratio = 0.8 to 1.0), plant indicators, critical growth stages (crown root, flowering, grain filling).

> Tip:
> Warabandi and rotational supply deliver water in proportion to land - transparent and equitable, combine with volumetric charges to save water.

4. Conjunctive Use of Water

Conjunctive use: the coordinated and planned use of surface water and groundwater so that the total yield is greater than separate use, with minimum environmental damage.

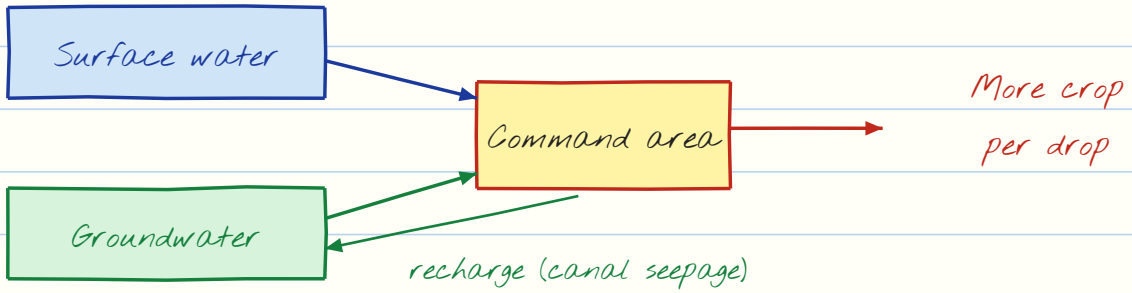


Fig 1: Conjunctive use idea

Why?

* Surface supply is seasonal, groundwater is available throughout the year but depends on recharge. Using both reduces risk.
* Canal irrigation causes waterlogging in head reach and shortage in tail reach; pumping lowers the water table (bio-drainage) and relieves waterlogging.
* Wells can supplement canal supply at tail and in droughts; canal water recharges the aquifer through seepage.

Methods

* Use surface water when plentiful (monsoon) and store excess in aquifer (artificial recharge); pump groundwater in lean season and drought.
* Blend saline groundwater with fresh canal water (cyclic or blending) for irrigation.
* Tubewell in canal command area (canal + well), tank + well (south India), conjunctive operation of surface reservoir and aquifer storage (Aquifer Storage and Recovery - ASR).
* Simulation and optimisation models (linear programming, MODFLOW + reservoir models) determine optimal pumping and surface supply.

Advantages and concerns

* Advantages: higher cropping intensity, more reliable supply, control of waterlogging and salinity, lower storage evaporation, saves canal capacity, cost-efficient.

* Concerns: over-draft if recharge is not planned, quality of groundwater (fluoride, arsenic, salinity), energy for pumping (subsidised power), institutional problem (two departments - one for canal and one for groundwater).

5. Participatory Irrigation Management (PIM)

PIM: involvement of water users in all aspects of irrigation management - planning, design, construction, operation, maintenance, monitoring and financing - by transferring responsibility to farmer organisations.

* Water Users Associations (WUA) at the water-course/minor level; Distributory Committee and Project Committee above it.
* Laws: Andhra Pradesh Farmers' Management of Irrigation Systems Act (1997) - first major reform, Maharashtra Management of Irrigation Systems by Farmers Act 2005; Pani Panchayats (Maharashtra, Odisha), Water Users Societies; Tamil Nadu WUA Act 2000; Joint Irrigation Management (JIM) - Gujarat Sinchai Mandali.
* Functions: distribute water equitably, collect water charges, maintain channels, resolve conflicts, plan cropping pattern, record keeping and share information.
* Benefits: efficient and equitable distribution, higher recovery of water charges, better maintenance, reduced conflicts, tail-end water reaching, improved yields and ownership.
* Constraints: lack of legal and financial powers, political interference, caste/land inequalities, weak training and technical support, poor condition of the physical system at hand-over.

Success:

PIM works when: users have legal rights and clear water entitlement, canal is in good condition, farmers get capacity building, and the government supports with funds.

6. Integrated Water Resources Management (IWRM)

IWRM (Global Water Partnership, 2000): a process which promotes the coordinated development and management of water, land and related resources in order to maximise economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems.

Principles

* Dublin principles (1992): (1) fresh water is finite and vulnerable resource; (2) development and management should be participatory; (3) women play a central role in provision, management and safeguarding; (4) water has an economic value and should be recognised as an economic good.
* Three E's: Efficiency (use water wisely), Equity (fair access for all), Environmental sustainability (leave water for ecosystems).

Elements (framework)

Pillar	Contents
Enabling environment	policies, laws, financing, national water plan
Institutional roles	river basin organisation, local bodies, WUAs, NGOs, private sector
Management instruments	water resource assessment, demand management, allocation, pricing, information, conflict resolution, regulation, monitoring

* River basin or watershed as the planning unit, link surface and groundwater, quantity and quality, upstream and downstream, land and water, green and blue water.
* Cross-sector integration (agriculture, industry, domestic, energy, environment) and integration across scales (local to national).
* In India: National Water Policy 2012, River Basin Organisations (Brahmaputra Board, Ganga basin authority), Water Resources Information System (India-WRIS), Jal Jeevan Mission, National Water Mission under NAPCC.

7. Water Management Policy During Droughts

Drought types: meteorological (rain deficit), hydrological (low streamflow/groundwater), agricultural (soil moisture deficit), socio-economic. IMD: drought if seasonal rain < 75 % of normal (rain deficiency > 25 %), severe if > 50 % deficiency.

Policy components

* Preparedness: drought monitoring and early warning (rain, SPI, NDVI, soil moisture, reservoir storage), vulnerability maps, contingency crop plans (CRIDA), drought manual 2016 (revised 2020).
* Water priority during drought: 1) drinking and domestic, 2) livestock, 3) protective irrigation for standing crops/perennial, 4) industry, 5) other. Cut allotment to low-value uses, ban water-intensive crops, restrict pumping and tanker supply in regions.
* Supply side: use dead storage with care, deepen wells and tanks, tanker supply, emergency recharge, water transport, desalination, treated waste water reuse.
* Demand side: rotational supply, volumetric limits, mulching, drip, water pricing, crop shift to pulses/millets, fallowing.
* Relief and livelihoods: MGNREGA work for water conservation, fodder banks and cattle camps, drinking water, crop and livestock insurance (PMFBY), seed supply, credit relief, public distribution.
* Long-term: drought-proofing - watershed development, tank desilting, groundwater recharge, farm ponds, afforestation, diversification, inter-basin transfer where required.

> Concept:
> Drought management cycle = prepare -> monitor -> respond -> recover -> mitigate. Shift from 'crisis management' (relief) to 'risk management' (preparedness).

8. Predicting Effect of Water Shortage on Crops

Crop yield falls when ET is less than potential (crop evapotranspiration ETc). The shortage effect depends on the growth stage and how severe the stress is.

$$(1 - Ya/Ym) = Ky\ (1 - ETa/ETm)$$

(Doorenbos & Kassam, FAO-33)

Ya = actual yield, Ym = maximum yield, ETa = actual ET, ETm = max ET, Ky = yield response factor (crop and growth stage specific).

Crop	Ky (whole season)	Critical stage
Maize	1.25	flowering / silking
Wheat	1.0 - 1.15	crown root, flowering
Rice	1.1 - 1.3	flowering, panicle initiation
Cotton	0.85	flowering, boll formation
Groundnut	0.70	flowering, pegging
Sugarcane	1.2	grand growth

* Ky > 1: sensitive (yield drops more than proportionally); Ky < 1: tolerant crop. Total shortage multi-stage product model: Ya/Ym = product of (1 - Kyi (1 - ETai/ETmi)) over stages.

Example

Maize, ETm = 500 mm, ETa = 400 mm, Ky = 1.25: 1 - Ya/Ym = 1.25 x (1 - 0.8) = 0.25 -> Ya = 75 % of Ym. A 20 % ET deficit gives a 25 % yield loss.

* Other approaches: crop growth simulation (AquaCrop of FAO, DSSAT, APSIM, InfoCrop) use soil water balance, canopy and harvest index to predict yield under deficit irrigation.
* Water stress indices: Crop Water Stress Index (CWSI) using canopy temperature, soil moisture depletion fraction p, relative water content, NDVI and remote sensing.

* Consequences: reduced germination, shorter growth, poor flower/grain fill, low biomass, quality loss; Strategy: deficit irrigation (RDI), supplemental irrigation at the critical stage, drought-tolerant varieties, mulching and conservation.

9. Water Footprint of Crops and Applications

Water Footprint (WF) - Hoekstra (2002): the total volume of fresh water used, directly and indirectly, to produce a product, measured over the full supply chain. For crops: m3 per tonne (or litre per Kg).

WF = Blue + Green + Grey (m3 per tonne of crop)

BLUE WF	GREEN WF	GREY WF
irrigation water (surface + ground)	rain water used by the crop (soil)	water to dilute pollutants

Water Footprint of a crop

Fig 2: Components of water footprint

$$WF_{crop} = (CWU_{blue} + CWU_{green}) / Y + WF_{grey}$$

Blue, green and grey water footprint

* Blue WF: surface and groundwater consumed (evaporated, incorporated in product, or not returned to the same catchment). For crops = irrigation ET (blue CWU) / yield.
* Green WF: rain water stored in the soil and used by the crop (rain-fed ET). Important for rainfed agriculture and forest.
* Grey WF: volume of fresh water required to dilute pollutants (fertiliser, pesticide) to meet ambient quality standards.

$$WF_{grey} = (\alpha \times AR) / (C_{max} - C_{nat}) / Y$$

(AR = chemical applied Kg/ha)

alpha = leaching/runoff fraction (e.g. 10 % for N), C_{max} = max acceptable concentration (e.g. 10 mg/L N), C_{nat} = natural conc.

* CWU: crop water use (m3/ha) = 10 x sum of ETc over the growing period (mm), Y = yield (t/ha). WF (m3/t) = CWU/Y.

Product (global avg)	Water footprint
Wheat	about 1800 m3 / tonne
Rice (paddy)	about 2500 m3 / tonne (India higher)
Sugarcane	about 200 m3 / tonne cane
Cotton lint	about 10,000 m3 / tonne
Beef	about 15,400 m3 / tonne

Applications

* Compare crops and regions; select crops with a low WF per Kg of food or per rupee and per calorie.
* Identify hotspots of water stress, evaluate sustainability (is blue WF within local environmental flow?), basin-level water accounting.
* Water-saving measures: increase yield (WF reduces), use drip, mulching, reduce fertiliser, treat waste water, improve irrigation efficiency.
* Virtual water trade: import of a water-intensive product saves local water - used in national water policy (e.g. rice, cotton exports from India).
* Product labelling, corporate water stewardship, national water footprint of consumption, policy for water-wise diets.
* Water Footprint Assessment 4 phases: set goals and scope, accounting, sustainability assessment, response formulation (Water Footprint Network manual).

Memory trick:

Blue = irrigation water, Green = rain stored in soil, Grey = pollution dilution water. WF reduces if yield goes up and losses go down.

10. Solved Examples and Revision

Example 1: Duty and delta

A crop with base period B = 120 days has duty D = 1200 ha/cumec on field. Find delta.

Delta = 8.64 B / D = 8.64 x 120 / 1200 = 0.864 m = 86.4 cm

Example 2: Irrigation requirement

ETc = 5.5 mm/day, effective rain = 0.5 mm/day, Ea = 70 %, Ec = 80 %, area 10 ha.

* Net irrigation = 5.5 - 0.5 = 5.0 mm/day
* Gross at field = 5.0 / 0.70 = 7.14 mm/day; at canal head = 7.14 / 0.8 = 8.9 mm/day
* Volume = 8.9 mm x 10 ha = 0.0089 m x 100,000 m2 = 890 m3/day

Example 3: Water footprint of rice

Green CWU = 4000 m3/ha, blue CWU = 8000 m3/ha, yield = 4 t/ha, N applied 100 Kg/ha, alpha = 0.1, Cmax = 10 mg/L, Cnat = 0.

* Green WF = 4000/4 = 1000 m3/t; Blue WF = 8000/4 = 2000 m3/t
* Grey WF = (0.1 x 100 Kg/ha) / (0.010 Kg/m3) / 4 t/ha = 1000/4 = 250 m3/t

Total WF = 1000 + 2000 + 250 = 3250 m3 per tonne

Quick revision list

* Rain water management: in situ + ex situ + recharge + safe disposal.
* Irrigation: GCA, CCA, duty, delta, efficiencies, warabandi, scheduling.
* Conjunctive use = surface + ground, planned together, avoid over-draft.
* PIM: WUAs with legal and financial powers; NRM: Dublin principles, efficiency-equity-sustainability.
* Drought: monitor, priority to drinking water, contingency plan, MGNREGA, insurance.
* Crop-water shortage: (1 - Ya/Ym) = Ky (1 - ETa/ETm).
* Water footprint: blue + green + grey.

Writing tip:

Exam approach: define -> draw a neat diagram -> write the formula with symbols -> give an example -> mention Indian case or policy.